Sentiment Analysis of Text Reviews

CSC 425 AI Project: Milestone 1 Proposal

Team Members & Responsibilities

Team: Solo project (1 member).

- **Obaid Alnaqbi** is involved in the entire project, which is a one-man team, for the problem definition, data collection/pre-processing, model development, model evaluation, documentation and presentation.

Communication & file sharing: As I make changes to my code, notes, and data, a single GitHub repository will be used that will serve as my Log of Progress and will be version controlled. I will be responsible for e-mailing the instructor.

Problem Description & Motivation

Sentiment analysis is the process of analyzing a text and categorizing the sentiment it contains, about a movie or product, as positive or negative. In this project, I want to create a system that will predict the sentiment of a review as positive or negative, based on the review.

It is one of the most common systems that I see on a daily basis, and here's why: I'm one of these people, before I watch a movie or buy something, I check the reviews. Big companies use something similar to collect customer sentiment about a product without having to read every single review themselves. There's also a lot of pre-existing labeled data for this project, so I don't need to build my own dataset from scratch, and because I already know the labels, I don't have to guess or make up my own.

Input, AI Task, and Expected Output

- **Input:** A short segment of unstructured text, e.g., a movie review or product review (a sentence to a few paragraphs).
- **AI Task:** Supervised binary text classification.
- **Output:** Predicted class of sentiment (positive or negative), optionally along with a predicted confidence level.

Dataset / Problem Environment

- **Source:** The data is provided by the Large Movie Review Dataset from IMDB (Maas et al., 2011) by the Stanford AI Lab.
- **Size:** The size of the labeled reviews is 50k (25k train, 25k test), and the ratio of positive to negative is good. If it is necessary to do an iteration early in the process, then a smaller subset of reviews (5,000-10,000 reviews) can be used.

- **Features:** Raw review text (after processing, this becomes tokenized words and word embeddings).
- **Sentiment Target (Output) Variable:** Binary sentiment label: positive (1) or negative (0).
- **Preprocessing needed:** Yes. The reviews will have to be preprocessed before I can use them for my application: removing stop words and punctuation, stripping out HTML tags, converting all the text to lower case, and splitting it into tokens. Then I'll turn those tokens into a form the model can read: TF-IDF for the baseline model, and embeddings for the neural model. I don't need to do the train/test split myself, since the dataset provided by IMDB is already split.

Proposed AI Approach

This is my first real NLP project, so I will start easy and then gradually work up to it:

1. **Baseline:** Train a Logistic Regression or Naive Bayes classifier using TF-IDF encoding of the reviews, with scikit-learn.
2. **Stretch goal (if time permits):** I'll see how well a small neural network and/or a small pretrained model (e.g., DistilBERT) performs compared to the baseline.
3. **Evaluation:** I will draw a confusion matrix and compute accuracy, precision, recall, and F1 score from the test set results of my model.

Initial References & Resources

1. Maas, A. L., et al. (2011). *Word Vectors Learning for Sentiment Analysis.* ACL. Introduces the IMDB dataset and the baseline sentiment-analysis techniques this project is based on.
2. Stanford AI Lab: IMDB Review Dataset page (ai.stanford.edu/~amaas/data/sentiment/). Official documentation and download for the dataset.
3. Scikit-learn documentation: "Working With Text Data" tutorial, a walkthrough of a TF-IDF and text-classifier pipeline.
4. Hugging Face `datasets` & `transformers` documentation, IMDB dataset card, for loading the IMDB dataset and, optionally, fine-tuning a pretrained transformer model.
5. Kaggle: "Bag of Words Meets Bags of Popcorn" competition page, a tutorial-style related project with worked-out examples and community notebooks on the same dataset/task.

Project Timeline

Period	Target
Approximately until Oct. 11	Formulate problem, pick dataset (IMDb movie reviews), set up GitHub repository
Oct. 11 – Oct. 25	Data exploration and preprocessing (cleaning, tokenization, TF-IDF); build baseline classifier (Logistic Regression / Naive Bayes)

Oct. 26 – Nov. 15	Expand the AI approach, tune baseline hyperparameters; implement and train the stretch-goal neural/transformer approach
Nov. 16 – Nov. 29	Review accuracy, precision, recall and the confusion matrix for both models; discuss errors and ways to fix them; improve the models
Nov. 30 – Dec. 11	Create report, demo, and presentation

Expected Final Deliverable

By the end of the semester, I will have a sentiment-analysis system (a package of Python code, in the GitHub repository) that will be able to take a review, such as a movie or product review, as input and output a prediction of whether the sentiment is positive or negative, along with:

- This includes a trained baseline model (and, if time allows, a comparison model based on a neural/transformer model),
- An evaluation report (accuracy, F1 metrics, and error analysis),
- A final written report and a brief live demo/presentation to show that the system works.